

Anti-Involution and Global Price Pressure: China's New Supply-Side Reform and the Spillovers of Green Industrial Overcapacity

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1 Introduction and the Policy Engine

China accounts for more than 70% of global electric-vehicle production (IEA, 2025a) and holds more than 80% of manufacturing capacity across key solar PV supply-chain stages. One policy apparatus built both positions, but the same apparatus has also created chronic overcapacity and price wars that now spill across borders.

From an economic perspective, China's phenomenon of "involution" is, in essence, a long-term imbalance between supply and demand caused by excess supply and insufficient demand, resulting in excessive, ineffective competition. Chinese-style involution is not caused by any single factor; rather, it is rooted in the interaction between China's institutional structure and its growth model. Market mechanisms are unable to correct this imbalance. Under conditions of excess supply and insufficient demand, the marginal returns on individual input decline, thereby triggering irrational internal depletion. My argument is that the anti-involution campaign Beijing announced in 2025 amounts to a supply-side reform *within* the industrial-policy state. The demand-side imbalance that makes overentry profitable goes untouched. So anti-involution disciplines the worst of the domestic price wars while the global economy absorbs the rest as downward pressure on traded-goods prices.

How does overcapacity get produced? The sequence starts with central-government target-setting. Made in China 2025 and the "new three" designation (electric vehicles, lithium-ion batteries, solar PV) tell local officials where state resources will flow. Competition for official promotion in China was based primarily on the GDP growth rate of the locality for which officials were responsible (Fang et al., 2025b). Under this system, the performance evaluation of local government officials was highly centered on the short-term goal of economic growth. In pursuit of maximizing political achievements during their terms of office, local governments competed with one another, actively intervened in the market, and provided large amounts of "gray subsidies" to enterprises within their jurisdictions. These "gray subsidies" include land concessions, loans at below-market interest rates, state-owned enterprise equity participation, and tax reductions, all used to attract investment and generate local GDP. Although China's central government has stated that GDP growth will no longer be used as the main criterion for official promotion, the endogenous motivation of local officials to pursue short-term growth is difficult to fundamentally change. The outcome is parallel investment in the same industries across dozens of provinces.

These gray subsidies distort the market mechanism's function of market selection. They have led to redundant construction and resource misallocation in many industries, causing resources to be allocated

to less efficient areas. In the past, Chinese local governments encouraged infrastructure investment in order to increase local GDP, resulting in serious overcapacity in industries such as steel and cement. Market fragmentation caused by China’s vertical and horizontal administrative segmentation compounds the problem further. State-controlled banks and local government financing vehicles keep lending for political reasons, so firms that should fail keep running. Branstetter and Li (2022) find exactly this for Made in China 2025 targets: subsidized firms invest more, but they do not consistently become more productive. Branstetter et al. (2023) confirm the pattern more broadly.

The last piece is the gap between what China produces and what its own consumers can absorb. China’s economy in recent years has entered what is known as the “new normal.” Growth has fallen from the double-digit rates of the past to around the official 5 percent target. Uncertainty about economic growth prospects, the withdrawal of foreign-invested enterprises, the closure of private firms, and young people’s lack of hope regarding employment and the future, have all weakened consumer confidence. Precautionary saving has risen and household spending has stagnated. China’s real estate market has continued to decline in recent years, with an increasing number of property developers experiencing debt crises. This has severely affected household balance sheets, as real estate is the largest form of household savings for Chinese residents. The shrinking of household wealth has directly weakened households’ consumption capacity. Due to insufficient confidence in the future economy, combined with the negative wealth effect caused by shrinking assets, consumer behavior has shifted toward downshifting in household consumption. At the same time, as local governments’ revenues from land finance have sharply declined, some localities have introduced various new fines or raised public service fees in order to increase revenue, indirectly pushing up the costs of manufacturing and transporting goods. Under the pressure of weak domestic demand and rising costs, enterprises can only be forced into price wars, and when the domestic side is structurally short, the surplus shows up as export pressure abroad.

Fang et al. (2025a) add that local governments systematically copy each other’s industrial policies, multiplying the problem across hundreds of cities. Rotunno and Ruta (2024) find these subsidies are associated with increased export volumes in the subsidized sectors. The “new three” are the obvious place to check.

2 Overcapacity in the “New Three”

Table 1 gives a snapshot of capacity pressure in the three sectors.

Solar PV. China’s solar PV manufacturing capacity exceeds 1 TW per year. Global installations in 2024 reached roughly 600 GW (IEA, 2025b). Even if every single panel installed on Earth were Chinese-made, nearly half the factory floor would be idle. Chinese firms hold more than 80% of global output at every major stage from polysilicon through module assembly. The OECD’s MAGIC database identifies solar as the most heavily subsidized sector it tracks: government support to solar firms averaged 3.2% of revenue, compared with 0.9% across all sectors (OECD, 2026, 2025).

Lithium-ion batteries. China hosts about 85% of global battery cell manufacturing capacity (USCC, 2025; IEA, 2025a). Batteries are an intermediate good whose demand depends on downstream adoption of EVs and grid storage. The bull case is that global battery demand will more than triple by 2030. But battery pack prices in China fell about 30% in 2024, against 10–15% in Europe and the United States

(IEA, 2025a). If this were a global learning curve, the price drops would look similar everywhere. Chinese prices fell roughly twice as fast, which looks more like a domestic glut than shared technological progress.

Electric vehicles. The EV sector is different again, and politically the most visible case. Because the Chinese government regards electric vehicles as an emerging industry in which China has the potential to gain global dominance, it has mobilized national resources to support the sector. According to estimates by the Center for Strategic and International Studies, from 2009 to 2023, China’s electric vehicle industry received approximately USD 230.9 billion in government subsidies. The main form of subsidy has been the vehicle purchase tax exemption policy, while direct consumer purchase subsidies were phased out at the end of 2022. Because automobiles are a consumer good with strong local-employment ties, provincial governments also have direct stakes in keeping regional automakers alive. Under these large-scale subsidies, the industry has become severely involuted, with wave after wave of price wars. Li Shufu, chairman of Geely Holding Group, criticized the industry as having “the highest degree of involution in the world.” NBS data show that automobile manufacturing PPI fell 2.8% in 2025, slightly worse than the 2.6% decline for industry as a whole (NBS, 2026). China’s domestic market cannot absorb the massive excess capacity driven by subsidies: nearly 20 million units of excess new energy vehicle capacity are expected by 2025, while global new energy vehicle sales in 2024 were only 17.76 million units and China’s market sales were 11.5 million units (IEA, 2025a; Mayr et al., 2026). The European Parliament study documents a rising share of loss-making firms in the automobile sector (Mayr et al., 2026). In China’s EV sector, the soft budget constraint described in Section 1 keeps loss-making firms alive, and the price floor that exit would establish never materializes.

Sector	China-side indicator	Demand benchmark	Price/margin symptom
Solar PV	Mfg. capacity >1 TW/yr	~600 GW installed (2024)	Sharp price decline
Li-ion batteries	~85% of global mfg. capacity	Below installed capacity	Pack prices fell ~30% in China (2024)
EVs	Capacity exceeds demand	Global sales >17M (2024)	Auto mfg. PPI –2.8% (2025)

Table 1: Descriptive indicators of capacity pressure in China’s “new three” sectors. Solar: IEA (2025b); OECD (2026). Batteries: USCC (2025); IEA (2025a). EVs: IEA (2025a); NBS (2026).

Is this really overcapacity, or forward-looking investment? Beijing says global green demand will eventually absorb the surplus. The OECD and IMF disagree, treating the gap as structural and rooted in the subsidy architecture of Section 1 (IMF, 2026; OECD, 2025). I side with the structural interpretation. Capacity investment follows the incentives of local officials and state lenders, not any firm’s assessment of future demand. Calling it “forward-looking” imposes a market rationality on a process that the tournament mechanism overrides.

In all three sectors, a fiscal and financial architecture prevents the surplus from correcting itself.

3 Anti-Involution as Domestic Response

The policy and its instruments

“Involution-style competition” (*neijuan shi jingzheng*) is regarded in China as a highly negative socio-cultural phenomenon. It refers specifically to a social predicament in which, within a closed space where resources are fixed or scarce, individuals are forced to participate in excessive and irrational competition,

ultimately forming a situation in which there is little hope yet no way to withdraw from the competition. In industrial terms, it means firms compete on price and scale alone, cutting margins without ever gaining a durable advantage. Every price cut gets matched, and the equilibrium is a race to the bottom where value is destroyed rather than created. In July 2024, Chinese authorities began to recognize the impact of involution on China's economy. At important meetings such as the Central Politburo meeting, the Central Economic Work Conference, the government's Two Sessions report, and meetings of the Central Financial and Economic Affairs Commission, they proposed measures such as "strengthening industry self-discipline to prevent involutionary vicious competition" and "comprehensively regulating involutionary competition and standardizing the behavior of local governments and enterprises."

Central authorities moved on two tracks. The first is administrative. On July 1, 2025, Xi Jinping chaired the sixth meeting of the Central Financial and Economic Affairs Commission, where he proposed further advancing the construction of a unified national market and lawfully regulating low-price, disorderly competition among enterprises, releasing the highest-level policy signal on anti-involution. A State Council meeting later that month on the new-energy vehicle industry called for curbing "irrational competition" and announced strengthened cost surveys and price monitoring across the NEV supply chain (State Council, 2025a). Earlier that year the NDRC issued guidelines for a unified national market, going after local protectionism and the regional barriers that fragment domestic demand and allow duplicated capacity to survive (State Council, 2025b). Other instruments include tighter safety and quality standards (revised EV standards take effect in 2026), merger facilitation, and a shift in how local officials get evaluated: less output quantity, more quality metrics.

The second track is more market-oriented, at least in design. On June 30, 2025, the Standing Committee of the National People's Congress passed the revised Anti-Unfair Competition Law, requiring that platform operators must not force or covertly force merchants on their platforms to follow their pricing rules and sell goods below cost, thereby disrupting market competition order. It also requires large enterprises not to abuse their advantages in capital, technology, transaction channels, industry influence, and other areas by forcing small and medium-sized enterprises to accept obviously unreasonable transaction terms, such as payment periods, payment methods, conditions, and liability for breach of contract. A concurrent Pricing Law overhaul aims to establish legal liability for predatory pricing. I think these provisions matter. If enforced, they would put a rule-of-law floor beneath pricing behavior that administrative instruments alone cannot provide. However, the Anti-Unfair Competition Law still does not confront the institutional roots that cause involution, namely the continued existence of local governments' "gray subsidies" and GDP-based assessment incentives. As long as this institutional distortion is not eliminated, local governments will continue to drive overcapacity, and vicious competition will be difficult to eradicate.

The MITI analogy and its limits

A serious counterargument says anti-involution is genuine policy learning, something like what Japan's MITI achieved. MITI used administrative guidance in the 1960s and 1970s to curb "excessive competition" (*kato kyoso*) in steel and automobiles, and the resulting consolidation contributed to technological upgrading (Johnson, 1982). So why not read anti-involution as a similar step? Japan's industrial consolidation coincided with one of the fastest sustained increases in household living standards in economic history. MITI could consolidate supply because domestic incomes were rising fast enough to make the consolidated firms profitable. China faces the opposite situation. Household consumption sits at 39–40% of GDP, well below advanced-economy norms (World Bank, 2025). The household savings rate is

around 34%, against a global average near 24% (Gunter et al., 2025). Klein and Pettis (2020) argue that financial repression and the channeling of income toward investment are structural features of the growth model. The growth model holds consumption down.

Anti-involution may slow the price war among EV manufacturers, but it cannot create the domestic buyers they need.

MITI also dealt with a handful of firms. China's overcapacity spans hundreds of cities and thousands of firms (Gunter et al., 2025). Beijing issues the directive, but the people who must carry it out are the same local officials whose budgets depend on the investment the directive is trying to restrain.

I read anti-involution as primarily a correction within the existing system. It combines administrative discipline with emerging legal instruments, but it leaves the demand side alone. Persistent involutionary competition harms corporate profitability and the healthy development of entire industries, causing large amounts of resources to be consumed in inefficient markets. Anti-involution may damp the worst price wars, but the underlying situation, production capacity exceeding domestic demand in a growth model that favors investment over consumption, stays as it is. Chinese companies have been forced to export domestic vicious involutionary competition into overseas markets, triggering an escalation of international trade protectionism and posing challenges to global economic stability.

4 Global Spillover: Downward Price Pressure

China Shock 2.0

When Autor et al. (2013) documented the original China Shock, the story was relatively simple: WTO accession accelerated a trade shock in labor-intensive manufacturing, and American workers in competing industries bore the cost. Two decades on, the shock has returned in a different form. What China now exports at scale is capital-intensive green technology, the “new three.”

China has genuine advantages in engineering talent and scale, but the policy apparatus from Section 1 amplified those advantages with directed capital, so the current shock layers policy-driven capacity expansion on top of real cost competitiveness. The twist is that cheap solar panels and batteries slow climate change, which makes the welfare calculus genuinely hard.

Strategic entry versus residual exports

Not all Chinese green-tech exports reflect the same logic. Some firms planned for export markets early, before domestic margins collapsed. BYD building plants in Thailand and Hungary is deliberate market entry. Other firms are redirecting output abroad because the domestic price war has left them unable to sell at home, closer to what economists call loss-tolerant export competition: margins below long-run average cost, sustained by soft budget constraints.

Evidence of downward price pressure

Chinese export volumes in clean-technology sectors have risen sharply since 2022 while unit values have fallen (USCC, 2025). Rotunno and Ruta (2024) provide econometric evidence that Chinese industrial subsidies are associated with higher export volumes and lower export prices in subsidized sectors.

A lot of commentary calls this “exported deflation,” which is the wrong framing. The decline is in the relative price of clean-technology goods, not in the general price level.

The green-public-goods counterargument

The strongest objection is that cheap solar panels and batteries speed up global decarbonization, especially in developing countries that cannot afford unsubsidized clean technology. For much of Sub-Saharan Africa and South Asia, affordable Chinese solar modules are the difference between decarbonized electrification and staying on fossil fuels.

Yet adjustment costs concentrate: workers in European solar-module assembly or American EV-component manufacturing bear losses while benefits spread thinly across consumers. The United States imposed a 100% tariff on Chinese EVs in 2024 (USTR, 2024). The European Union adopted countervailing duties, 17.0% for BYD up to 35.3% for SAIC (European Commission, 2024). No major economy has taken the subsidized goods and compensated displaced workers at home. The welfare benefit of subsidized overcapacity requires open markets, and overcapacity is closing them.

5 Tensions and Outlook

Anti-involution targets the wrong end of the problem. The shortfall in domestic demand is the deeper issue, and quality thresholds and merger facilitation do nothing about that.

On the domestic front, the most likely trajectory is a muddled middle. Solar manufacturing may see some rationalization as small firms exit, but the EV sector is harder to consolidate because local governments hold stakes in regional automakers and no official wants to shut a factory that employs constituents.

The more interesting adjustment channel is overseas. Chinese EV supply-chain firms invested more abroad than at home for the first time in 2024 (IEA, 2025a), and industry estimates put overseas manufacturing capacity roughly doubling by 2026. Part of this is tariff-driven: companies are localizing production to get around trade barriers. But if Chinese firms relocate a meaningful share of their capacity to Southeast Asia and Latin America, they could endogenously relieve some of the domestic overcapacity that anti-involution was supposed to address.

But relocating factories is not the same as fixing the underlying problem. The fiscal incentives inside China that push local officials to build capacity have not changed at all.

If involution remains unresolved domestically, the consequences extend beyond trade. Persistent involutionary competition erodes firm profitability and drags down aggregate productivity, creating a vicious cycle in which weak growth reinforces the conditions for further involution. Youth labour-market prospects deteriorate, reducing social mobility and feeding the disengagement that Chinese commentators call “lying flat.” And by compressing margins across entire sectors, involution discourages the risk-taking that industrial upgrading requires.

Genuine structural reform would mean raising the household share of national income: expanded social insurance, fiscal restructuring away from land-sale dependence, relaxation of financial repression. If the government invests excessive resources in “dividing the pie well” but the speed of “making the pie bigger” cannot keep pace, China may ultimately face the risk of falling into “common poverty.” Long-term weak domestic demand may become normalized, making China’s attempt to counter external pressure by “strengthening the domestic circulation” highly challenging. Anti-involution does not ask any of these questions.

The precedents are not encouraging. China has been talking about rebalancing toward consumption since the 12th Five-Year Plan (2011–2015), and the household consumption share of GDP has barely moved. What I see is a Chinese industrial sector that has learned to work around the constraints it faces, finding new outlets for overcapacity while leaving the underlying causes in place.

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